Problem	Penalty	Free	Informed	Best penalty	Best free	Best informed	COLSHADE	L-SHADE Deb	L-SHADE pen.	L-SHADE ε	L-SHADE ε-long	CMA-ES Deb	CMA-ES pen.	Random
lens/original	0.611 ± 0.029	**0.692 ± 0.022**	0.641 ± 0.054	0.638 ± 0.030	0.695 ± 0.023	**0.714 ± 0.029**	0.582 ± 0.012	0.586 ± 0.014	0.585 ± 0.014	0.586 ± 0.011	0.625 ± 0.008	0.562 ± 0.017	0.556 ± 0.014	0.555 ± 0.017
lens/realistic_draft	0.662 ± 0.040	**0.675 ± 0.042**	0.647 ± 0.039	0.697 ± 0.048	**0.706 ± 0.034**	0.668 ± 0.028	0.579 ± 0.010	0.582 ± 0.011	0.587 ± 0.013	0.587 ± 0.015	0.620 ± 0.010	0.565 ± 0.015	0.555 ± 0.016	0.551 ± 0.017
cec/RC12	0.516 ± 0.014	0.718 ± 0.113	**0.732 ± 0.106**	0.518 ± 0.015	**0.794 ± 0.093**	0.788 ± 0.095	0.643 ± 0.017	0.621 ± 0.018	0.512 ± 0.016	0.630 ± 0.014	0.583 ± 0.016	0.505 ± 0.007	0.490 ± 0.011	0.372 ± 0.029
cec/RC13	**0.963 ± 0.030**	0.955 ± 0.021	0.953 ± 0.014	**0.979 ± 0.005**	0.954 ± 0.031	0.961 ± 0.009	0.849 ± 0.010	0.795 ± 0.010	0.799 ± 0.008	0.798 ± 0.009	0.758 ± 0.015	0.655 ± 0.013	0.661 ± 0.015	0.605 ± 0.014
cec/RC15	0.940 ± 0.034	**0.960 ± 0.013**	0.953 ± 0.013	**0.976 ± 0.005**	0.971 ± 0.006	0.964 ± 0.003	0.711 ± 0.004	0.696 ± 0.006	0.699 ± 0.005	0.695 ± 0.006	0.680 ± 0.007	0.603 ± 0.010	0.605 ± 0.008	0.539 ± 0.021
cec/RC16	**0.563 ± 0.081**	0.561 ± 0.085	0.516 ± 0.051	0.623 ± 0.118	**0.656 ± 0.086**	0.558 ± 0.087	0.486 ± 0.005	0.496 ± 0.005	0.494 ± 0.006	0.494 ± 0.004	0.493 ± 0.003	0.454 ± 0.014	0.423 ± 0.028	0.418 ± 0.020
cec/RC18	**0.648 ± 0.092**	0.622 ± 0.066	0.635 ± 0.077	**0.698 ± 0.100**	0.648 ± 0.079	0.682 ± 0.093	0.591 ± 0.015	0.606 ± 0.025	0.645 ± 0.027	0.606 ± 0.025	0.622 ± 0.018	0.515 ± 0.008	0.539 ± 0.014	0.506 ± 0.007
cec/RC21	**0.935 ± 0.070**	0.885 ± 0.112	0.926 ± 0.067	**0.939 ± 0.054**	0.937 ± 0.038	0.917 ± 0.077	0.811 ± 0.016	0.743 ± 0.014	0.738 ± 0.010	0.743 ± 0.014	0.729 ± 0.008	0.611 ± 0.014	0.612 ± 0.018	0.540 ± 0.014
cec/RC25	0.511 ± 0.017	**0.535 ± 0.028**	0.529 ± 0.020	0.508 ± 0.014	**0.561 ± 0.040**	0.535 ± 0.020	0.515 ± 0.010	0.514 ± 0.007	0.506 ± 0.009	0.511 ± 0.007	0.510 ± 0.009	0.497 ± 0.002	0.496 ± 0.004	0.494 ± 0.004
cec/RC26	**0.557 ± 0.042**	0.518 ± 0.045	0.513 ± 0.029	**0.580 ± 0.044**	0.520 ± 0.032	0.523 ± 0.043	0.375 ± 0.011	0.427 ± 0.011	0.446 ± 0.015	0.430 ± 0.019	0.418 ± 0.011	0.505 ± 0.016	0.547 ± 0.026	0.144 ± 0.017
Mean	0.691 ± 0.016	**0.712 ± 0.018**	0.705 ± 0.017	0.716 ± 0.016	**0.744 ± 0.017**	0.731 ± 0.012	0.614 ± 0.004	0.607 ± 0.004	0.601 ± 0.004	0.608 ± 0.004	0.604 ± 0.003	0.547 ± 0.004	0.548 ± 0.007	0.472 ± 0.006

Test AOCC (higher is better), mean ± std over test runs: LLaMEA arm = best algorithm of each of the 5 LLaMEA runs × 15 seeds (75 values); baselines = 15 seeds.

Best: the arm's best algorithm, chosen by training fitness, over its 15 test seeds.

Mean row: per test run, the mean over the problems. Bold: best of arms and baselines; in the Best block, best of the three.